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Crime Prediction Using Spatio-Temporal Data

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Abstract. A crime is an action which constitutes a punishable offence by law. It is harmful for society so as to prevent the *criminal activity*, it is important to understand crime. Data driven researches are useful to prevent and solve crime. Recent research shows that 50% of the crimes are committed by only handful of offenders. The law enforcement officers need early information about the *criminal activity* to response and solve the spatio-temporal criminal activity. In this research, supervised learning algorithms are used to predict *criminal activity*. The proposed data driven system predicts crimes by analyzing San Francisco city criminal activity data set for 12 years. Decision tree and k-nearest neighbor (KNN) algorithms are applied to predict crime. But these two algorithms provide low accuracy in prediction. Then, random forest is applied as an ensemble methods and adaboost is used as a boosting method to increase the accuracy of prediction. However, log-loss is used to measure the performance of classifiers by penalizing false classifications. As the dataset contains highly class imbalance problems, a random undersampling method for random forest algorithm gives the best accuracy. The final accuracy is 99.16% with 0.17% log loss.

Keywords: Crime prediction · Crime · Machine learning · Spatio-temporal activity · Data analytics · Ensemble methods

1 Introduction

Criminal activities are present in every aspect of human life in the world. It has direct impact on the quality of life as well as on the socio-economic development of a nation. As it has become a major concern almost for all countries, governments are prone to use advance technologies to tackle it. Crime Analysis is a sub brunch of criminology which studies the pattern of the crime and criminal behavior and tries to identify the insinuations of such event. A machine learning agent typically uses dataset for predictive analysis where it employs different techniques to find patterns [18, 20, 26]. Besides these, a receny-based analysis and corresponding patterns are also effective in predictive analytics [19]. A machine learning agent can learn about a crime and analyze the pattern

of occurrence of a crime based on the previous reports of criminal activities, and thus find the crime hotspots based on time, type and other factors. This technique is known as classification problem, and it allows predicting nominal class labels [32]. Classification techniques are used in many different domains such as financial market, business intelligence, healthcare, weather forecasting etc. [17]. Law enforcement agencies use analytical information, and employs different patrolling strategies to keep an area secure. In this paper, dataset from San-Francisco open data [1] is used which contains records of criminal activities for the last twelve years. It uses classification techniques such as decision tree, KNN as well as ensemble methods like random forest, adaboost that are well-known in the area of machine learning [25], to find hotspots of crime zone for specific time of day. Later, the results of algorithms are compared, and the most effective approach also registered for future use.

Criminal law and sociology scholars studied the pattern of crime and its relation to socio-economic development. Studies show that notable amount of crimes committed in the micro level of a region. These clusters are known as hotspots. Researchers claim that a good neighborhood often has few streets or certain areas where higher amount of criminal activities happened compared to other areas. Geographical topology and micro structures are more important for dealing with hotspots. Machine learning can implement the same theory by taking data driven approach to identify hotspots [31]. The main goal of this research is to use various machine learning algorithms to find the clusters, and analyze the previously reported criminal activities to discover hidden patterns. Law enforcement agencies can deal with crimes effectively if they have prior hints about crime activities and information about crime hotspots.

Crimes are being treated as spatio-temporal events. These events can be explained in term of space, time, and attributes where space specifies the location of the crime and time describes the time instant and attributes specify occurred events. Finding spatio-temporal patterns of crime help us to explain the association among crimes, space, and time. Machine learning approaches focus on the analysis of a single attributes such as robberies or on a specific pattern such as crime hotspots [29]. Spatio-temporal outliers, hotspots, coupling, and partitioning are very crucial for crime prediction. Spatio-temporal outliers are events whose attributes varies significantly from other objects in its neighborhood and are used to find unexpected events (e.g., crime). Spatio-temporal coupling occurs in close spatio-temporal proximity. For example, frequent occurrences of robberies occur close to a bar and supermarket on weekends. Spatio-temporal clustering means grouping similar crimes based on time and space. It is used to discover the hidden pattern of criminal activities. Spatio-temporal hotspots are places where the number of events is extremely high. It helps law enforcement organization to detect and limits the criminal activities [13].

2 Related Research

Researches in criminal activities prediction are given priority in all over the world. Most of the researches study relationship among criminal activities, poverty and socio-economic variables like income level [15], unemployment [8], race [4], and level of education [7]. Crime hotspots are located by analyzing demographic data and mobile

networks in London [3]. They have implied that anonymized data, collected by mobile networks, contain indicators for predicting crime levels. It combines two data sets – i) 1990 US LEMAS and ii) crime data 1995 FBI UCR, and then applies traditional classification algorithms like naive Bayesian algorithm and decision tree classifier. Thus, 83.95% accuracy has been achieved for predicting a crime category of various areas in USA [9]. However, this paper fails to explain imbalanced classes of crime category, if any. Somayeh et al. [28] used some advanced machine learning techniques by using same databases, where KNN algorithm prediction accuracy is higher (89.50%) than other machine learning techniques. They also used chi-square test to improve the feature selection. Wang et al. [30] proposed a different machine learning technique known as series finder. It find the underlying patterns of criminal activities committed by same person or group of persons.

Clustering is also another common technique used to predict patterns of geographic criminal history and behavior [8]. Remond and Baveja [16] proposed case-based reasoning (CBR) system which filtered out police cases for better prediction. It worked on noisy data, and explained how police reports show only individual criminal activities and do not addresses gang crimes. Social networks are also used as a potential source of criminal activity indicators. Sadhana and Sangareddy [10] used twitter information to find real time criminal offences. They also used the same data set to find the concentration of crime occurrences, and find large scale of hotspots. Thus several techniques [6, 11, 12, 14, 27] are used for analyzing crimes.

3 Dataset Analysis

3.1 Crime Dataset and Attributes

The experiment is conducted on a specific dataset. The dataset is provided by San Francisco (SF) open dataset from SFPD crime incident reporting system [9].

It provides information about crime incidents that occurred in San Francisco from 1/1/2003 to 5/13/2015. The dataset is in a csv file which contains 8,78,049 rows. The

Table 1. Attributes of the crime dataset.

Date time	A timestamp of crime occurrence
Category	Type of crimes. This is the target label for the data. There are 39 types of crimes listed in the data set Crime Description A detailed description of specific type of crimes
Day	Day of week
pDistrict	Police department's district name. There are total 10 police districts in the data
Resolution	How the crime was solved.17 types of resolution are stated
Address	Incident/occurrence street address
X	It signifies the latitude of the location of the crime
Y	It signifies the longitude of the location of the crime

features and label can be determined from the list of attributes in Table 1. The target label needs to be predicted is the category of a crime incident. The attributes crime description and resolution are also related to the target label. Hence, all other attributes apart from these three attributes are used as features.

Table 2. Attributes of the crime dataset.

Category	Frequency
Theft of property	174900
Other offences	126180
Non-criminal	92300
Physical assault	76876
Illegal drug	53970
Theft of vehicle	53781
Damage property	44725
Warrants	42214
Illegal entry	36755
Suspicious activity	31414
Kidnappings	25989
Robbery	23000
Fraud	16679
Forged	10609
Secondary codes	9985
.....	
Trespassing	6

There are 39 types of crimes in the San Francisco crime dataset shown in Table 2. These types are considered as classes. Hence, 39 classes made San Francisco a multi class classification problem. There are few crimes that occur very frequently, and some crimes are really rare. Larceny/Theft is the most common crime with a frequency of 174900, and Trespassing (TREA) is the least common crime with a frequency of 6. There are 14 crime classes that occurred more than 10,000 times and 14 crime classes occurred less than 2,000 times. It resembles that the classes are not evenly distributed.

3.2 Features

From date time stamp, four main features are extracted which are year, month, date, hour.

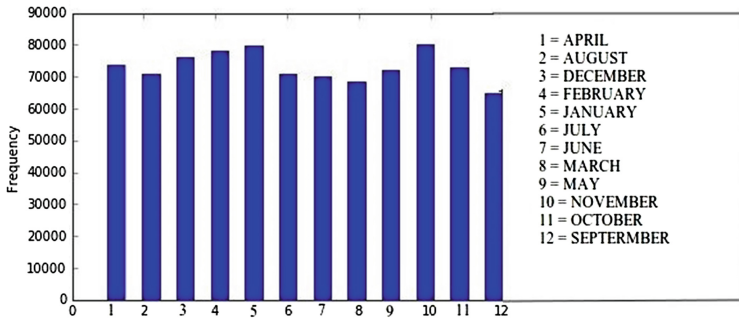


Fig. 1. Crimes in different months

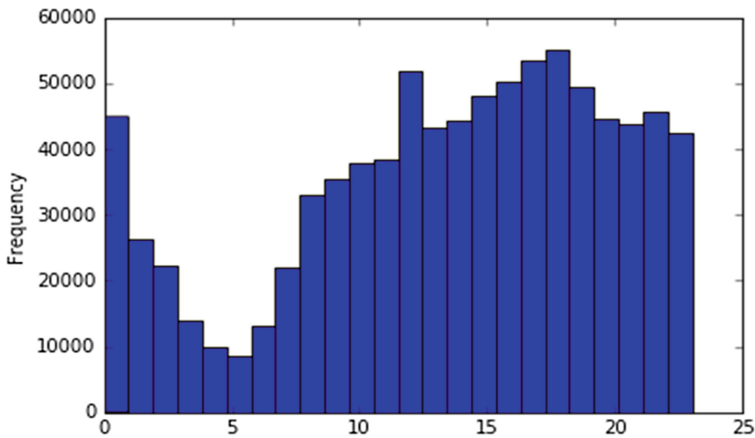


Fig. 2. Crimes in different days

The plots shown in Figs. 1, 2, and 3 indicate the criminal activities occurring throughout of the year. It is observed that summer and winter have less criminal activities compared to other seasons. Most of the crimes occur on Friday, where least crimes occur on Sunday. Crime rates almost gradually increase from Monday to Thursday. This however does not seem indicative of any pattern. The time of crime occurrence depicts an interesting picture (Fig. 4).

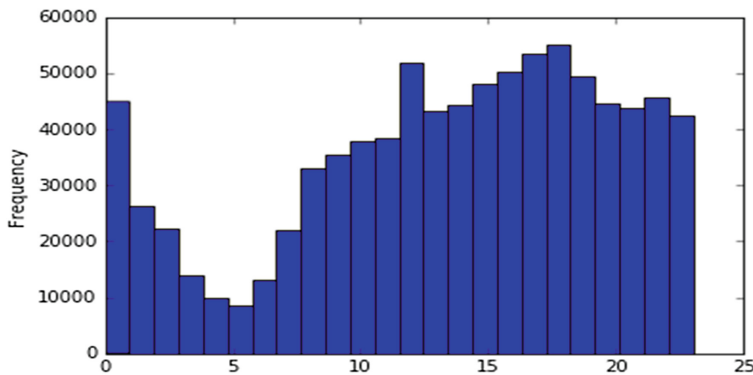


Fig. 3. Crimes in different hours

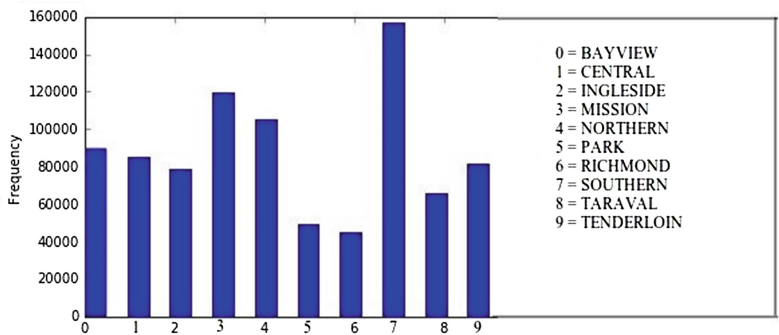


Fig. 4. Crimes in different police districts

Most crimes occur during afternoon to evening where as it is low at midnight to morning. There is an upsurge of criminal activities at 6 PM and 8 PM. Criminal activities are drastically reported around 9 AM, and it continues to show a gradual increase throughout the day, peaking at 6 PM. Among the ten police districts, criminal activities in the southern district are higher than any other districts.

4 Methodology

4.1 Preprocessing

Python library scikit-learn (sklearn) is used for preprocessing the dataset. Some attributes in the csv files contain string values as well as numeric values. In order to use this dataset in machine learning models, the text features need to be converted into a numeric value. Python library numpy is used to convert string into numeric values. Attributes with string data type are “Day”, “Category”, “Address” columns. Scikit-learn have a preprocessing package that converts string data into numeric data. This package gives an integer value to each unique item after sorting items in ascending alphabetical order. Date-time attribute

is also a string data type which is converted into a date-time object, and four different attributes such as “Hour”, “Date”, “Month” and “Year” are obtained from it.

4.2 Training and Testing Dataset

To avoid over fitting, and getting more realistic accuracy, the dataset is divided into two portions: testing dataset and training dataset. Training dataset contains all features along with the target label. Testing dataset only contains the features from which a machine learning model predicts the target label. Scikit-learn’s model selection package contains a class `test train split` that splits the original dataset into testing and training dataset. The default value of the test dataset size is 25% of the original dataset. This default value is used in the conducted experiments.

4.3 Feature Extraction and Selection

While the given features give sufficient information about a crime incident, new features can be extracted from the given features which might be proven to be useful. Analysis of different features shows that there is a specific pattern in occurrence of crime during different parts of a day. A new feature can be extracted by dividing a day in few different parts rather than considering 24 h. Although, in [22, 23] the authors presented a dynamic time segmentation technique, we consider a static segmentation based on our crime data. Figure 3 shows that crimes start falling after midnight, and rise gradually from afternoon to evening. Therefore, a day can be divided into following parts:

- Early morning: 1 AM–7 AM.
- Late morning: 8 AM–1 PM
- Afternoon: 2 PM–7 PM
- Night: 8 PM–12 AM

With blocks of time introduction, a clearer picture of time can be found which performs better than 24 hour-cycle-time. With blocks of time, a linear data is achieved compared to an arbitrary sequence of occurrence of criminal activities that does not reflect any indication. Principal component analysis (PCA) is a method of linear dimensionality reduction. It projects data in a lower dimensional space and maximizes variances. PCA class of `sklearn.decomposition` package is used to obtaining variable features from the existing 9 features. An `N` component in the PCA class indicates the number of new features with lower dimension to be extracted from the old features.

After preprocessing, total number of features is 9. One assumption is made before using any predictive model is that some features might be more useful than others. Too many features can make classification complicated and cause overfitting while more features describe the data better. `Sklearn.feature selection` module uses univariate statistical tests to find features that are best related to the target label. It can select percentile class of this module takes a percentage input, and returns that percentage of best features. Different percentages of features are used in different models to see which arrangement features gives better performance.

5 Result Analysis

We have used supervised classification method for this experiment. Performances of different machine learning models are discussed below:

5.1 Decision Tree

Decision tree classifier can solve both regression and classification problems. It has many parameters like `n_estimators`, `criterion`, `max_depth`, `min_samples_split`, `max_features`, `max leaf nodes`, `n jobs`, `random state`, `verbose`, `class weight` but only two parameters are convenient to use in this case. The number of splits can be made by decision tree, and indicated by `min_samples_split` and `and`. The function is to measure the quality of a split, is indicated by `criterion`. Information gain and impurity are the two types of functions which are used for measuring quality of split [21].

Table 3. Classification result for different parameters of Decision Tree.

Split	Function	Accuracy	Log-loss
50	gini	28.26%	8.41
100	gini	29.80%	5.45
300	gini	30.72%	3.31
500	gini	30.45%	2.83
50	entropy	29.24%	8.41
100	entropy	30.43%	5.52
300	entropy	31.17%	3.31
500	entropy	30.56%	2.91
600	entropy	30.46%	2.76

We considered the criteria “gini” for the gini impurity and “entropy” for the information gain [24]. In this case, four columns - split, function, accuracy, and log-loss are used for showing the output in Table 3. The best accuracy is 31.17% when log loss is 3.31, function is entropy, and 300 split. The lowest accuracy is 28.26% when log loss is 8.41%, function is gini, and 300 split. It is observed that split and function have very little impact on accuracy.

5.2 K-Nearest Neighbor

Sklearn.neighbors offers *KNeighborsClassifier* function. KNN has many different parameters like `n_neighbors`, `metric`, `metric_params`, `weights`, `algorithm`, `leaf_size`, `p`, `n_jobs` but `metric` and `n_neighbors` are very useful among these parameters. Here parameter `n_neighbor = 50` is giving us best accuracy score of 28.50% when log loss is 5.04. The lowest accuracy score is 27.91% when `n_neighbors = 500`, and log loss is 2.62. This shows that log loss is high when accuracy is low. Now, we use feature selection methods to find out whether the accuracy can be improved or not (Table 4).

Table 4. KNN result for different neighbor value.

Features	Log n neighbor	Accuracy (%)	Log Loss
All	30	28.14	6.61
All	50	28.50	5.04
All	70	28.39	4.26
All	100	28.41	3.71
All	200	28.35	3.02
All	300	28.15	2.78
All	400	27.96	2.69
All	500	27.91	2.62

5.3 Adaboost

In adaboost, *base_estimator* and *n_estimator* are two parameters, where *n_estimator* means the number of weak classifier that is used in boosting, and *base_estimator* is used to terminate the boosting. We are going to boost the decision tree result through adaboost ensemble method. Adaboost classifier shows same log loss value but reduces misclassification. Accuracy measurement shows that when we used 100 estimators for decision tree it gives very low accuracy (8.80%) with the log loss 3.10. However, the accuracy is improved when we used the estimator value 10. Predictive ability for both estimator values remains same (Table 5).

Table 5. Result for Adaboost.

Number of trees	Accuracy	Log loss
10	31.22	2.34
50	31.70	2.28
100	31.71	2.28

5.4 Random Forest

Random forest classifier also has various parameters like decision tree but only *min_samples_split*, *criterion*, and *n_estimators* parameters are used in this paper. We used *n_estimators* to find the number of trees in the forest, *criterion* indicates the function which measures the quality of a split, and *min_samples_split* indicates minimum number of samples which is required to split an internal node. Both accuracy and log loss are improved while the numbers of trees in random forest are increased. Ten, 50, and 200 estimator gives accuracy of 31.22%, 31.70%, 31.71% and log loss of 2.34%, 2.28, 2 respectively (Table 6).

Table 6. Random forest’s result.

Number of trees	Accuracy	Log loss
10	31.22	2.34
50	31.70	2.28
100	31.71	2.28

5.5 Oversampling Dataset

Oversampling method makes the dataset more balanced by synthesizing new minority class samples [2]. Many functions are available to balance an imbalanced class using oversampling method. In this research, we use SMOTE oversampling technique. In SMOTE, KNN are used for generating synthetic minority classes by operation on feature space. Here, value of k relies on the number of samples need to be created [2].

5.6 Undersampling Dataset

Undersampling is applied to the majority classes in an imbalanced dataset. It under-sampled the majority classes by compensating minority classes [5]. It tells the machine learning agents not to become biased and not to ignore the false positives. We use *imblearn* function from *scikit-learn* package to do the undersampling in the imbalance data.

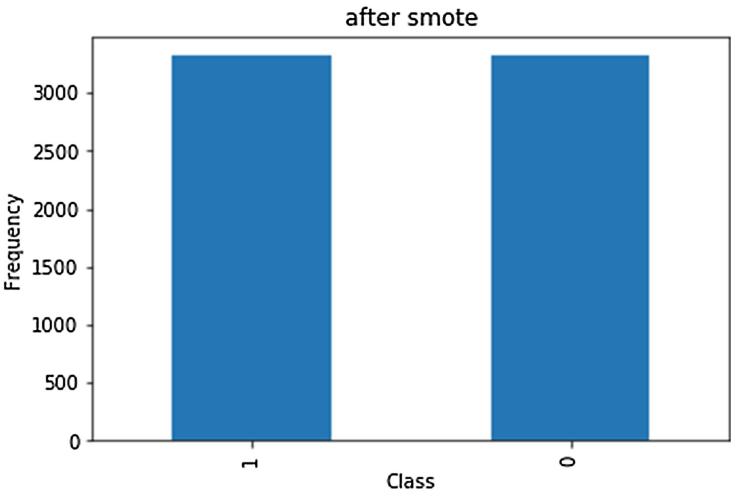


Fig. 5. Class frequencies after random undersampling.

5.7 Random Undersampling

Random undersampling is an efficient and effective technique to balance the data by randomly selecting a subset of data for the targeted classes [5]. Here, random undersampling reduces the frequency of majority class when considering minority class frequency. As a result, classes become balanced, hence same models and parameters can be used. SMOTE oversampling and random undersampling methods are used to boost the result, and made the class more balanced. Figure 5 shows that both classes are balance in the best possible manner. All the classifiers give a better result after introducing oversampling and undersampling methods. Results are improved by a very good margin. After resampling with random undersampling, random forest shows 99.16% accuracy score along with the log loss value of 0.17. Finally, best accuracy score is obtained after applying undersampling (Table 7).

Table 7. Classification result after ENN undersampling.

Sampling	Method	Models	Accuracy	Log loss
Over sampling	SMOTE	Random forest (num of tree = 100)	73.89	0.58
Undersampling	Random undersampling	Random forest (num of tree = 100)	99.16	0.17

6 Conclusion

Machine learning agent can classify a criminal activity using basic details of a crime occurred in an area with time and location. San Francisco dataset has 39 classes, and frequencies of all classes were not equally distributed. As the classes were poorly imbalanced, machine learning agent failed to perform well in the original dataset. From the original dataset, machine learning agents managed to provide a poor accuracy score of 31.71% that is pretty low. So we divided the 39 classes into two classes. One is frequent class and other one is rare class. The frequent class consists of most frequent crimes, and the rare one consists of least frequent crimes. As we expected, machine learning agents performed well in remodeled dataset and resulted accuracy is 68.03%. To overcome the imbalanced problem, we used oversampling and undersampling methods. Machine learning agents can be highly benefited after using these two methods. With an accuracy of 99.16%, random forest performed the best decision making classifier than other machine learning algorithms.

We like to improve our crime prediction accuracy in future. Currently, we use few machine learning techniques like decision tree, KNN, adaboost, and random forest. We plan to use deep learning technique to improve prediction result. In addition, we try to incorporate cybercrime prediction with the real world crime prediction.

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